Harish Manoharan

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Software engineer at Purgo AI since 2025: retrieval, verification and evals for an LLM agent. M.S. CS, UT Dallas, 2026.

EXPERIENCE

Purgo AI San Francisco Bay Area
AI startup: an agent that turns data-engineering tickets into PySpark, SQL and dbt code
Software Engineer Jun 2026 – Present
Software Engineer Intern (on site, then remote during M.S.) Jun 2025 – May 2026

- Cut missed source tables from 79 to 24 on our 103-ticket internal benchmark by adding dbt lineage expansion (SQL parsing with sqlglot) to the LangGraph agent's retrieval, with no ticket losing a table it had already found.
- Built the Polaris/Iceberg catalog search behind the agent's retrieval (Pinecone, keyword fallback): recall@10 0.68 vs 0.54 and 0.18 s vs 4.0 s against Databricks Vector Search on 188 queries, plus PII/PHI masking (Presidio) on sampled rows.
- Turned recurring Databricks runtime failures into AST lint rules that the agent's self-review runs on generated PySpark/SQL, plus a recovery pass that keeps the best-scoring candidate when all fail.
- Rewrote the retrieval and drafting prompts for a smaller reasoning model with per-stage reasoning effort, switching only after 5 runs on 102 tickets showed recall held within noise; those stages (about 6% of run cost) are now 7.6x cheaper.
- Built the compliance-validation (IQ/OQ) engine that checks life-sciences customers' Databricks platforms on Azure and AWS (SigV4): it runs a git-versioned library of 59 tests as cancellable jobs with per-test timeouts, keeps redacted evidence for signed audit reports, and turns requirement documents into new tests with an LLM.
- Built the LLM validation product across UI, API and test engine (React/Next.js, NestJS, PostgreSQL): 10 behavioral tests on Azure OpenAI or Databricks-served models (hallucination on missing fields, extraction F1, demographic bias, 20-run determinism, latency at 20 concurrent calls) and 9 checks on an Azure deployment's access, network and audit setup.

Digitus Business Solutions Bengaluru, India
Software Engineer Intern May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building the text-to-SQL agent behind a database chatbot used mainly by the sales team, with schema discovery and retries driven by execution errors.
- Cut median time to resolution from 20 to 6 minutes with a multimodal RAG assistant over slide decks and videos (enforced citations). Also built a QAD ERP to MyInvois e-invoicing integration.

PwC India Bengaluru, India
Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Cut forecast error (MAPE) 15% against the baseline across 50+ SKUs with XGBoost + SARIMA forecasts in Power BI.

EDUCATION

The University of Texas at Dallas M.S. Computer Science GPA 3.9/4.0 · Dean's Graduate Scholar May 2026
Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences May 2024
INSPIRE Scholar · Young Research Fellow · Thesis (The Jackson Laboratory): unsupervised behavior discovery in mice

PROJECTS

- **ReadableSTEM** (readablestem.com, open source, live): turns STEM course PDFs, even scanned or handwritten ones, into accessible HTML with MathML. Each equation is checked against the PDF's text and escalated from a cheap to the main model, then to instructor review, under per-job spend caps. Evaluated on two pre-registered held-out sets (textbooks, arXiv papers), published where a chatbot does better.
- **Margins** (studymargins.com, open source, live): turns a student's course files into cited notes, quizzes, practice exams and a day-by-day plan to the exam date (Claude, hybrid retrieval on pgvector).

SKILLS

LLMs: RAG, evals, AI agents (LangGraph), PyTorch · **Cloud and tools:** AWS, Azure, Docker, Git, GitHub Actions
Data: Databricks, Apache Spark (PySpark), dbt, sqlglot, PostgreSQL, pgvector, Apache Iceberg, Pinecone
Languages and web: Python, TypeScript, JavaScript, SQL, FastAPI, NestJS, React, Next.js